Inventory Adjustment Workflow

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> Fictional validation document. All entities, codes, dates, and values are invented; no production or personal data.

1. Objective and scope

The Inventory Adjustment workflow records controlled changes to stock balances in a fictional warehouse. It supports damage, cycle-count correction, and approved write-off. It does not perform procurement or vendor settlement.

2. Requirement matrix

Key	Capability	Acceptance rule	Severity
INV-01	Create adjustment	Require warehouse, SKU, quantity delta, reason, and requester. Reject a zero delta.	Must
INV-02	Approval threshold	Absolute value of quantity delta above 25 units requires a second approver.	Must
INV-03	Negative inventory	Block an adjustment that would reduce available stock below zero unless the reason is STOCKTAKE_CORRECTION and a supervisor approves.	Must
INV-04	Optimistic locking	Reject stale updates with INV-409 and return the latest stock version.	Must
INV-05	Audit history	Store before quantity, after quantity, actor, decision, and event time.	Should
INV-06	Bulk upload	Accept up to 250 rows; each invalid row includes a row number and reason.	Should

3. Operator procedure

Operators follow the sequence below. Each numbered instruction is a separate step, not a two-column table.

1. Open Stock Control and select the target warehouse.
2. Search by SKU or scan the warehouse label.
3. Enter the difference, select a reason, and provide supporting notes when required.
4. Submit for approval if the configured threshold is exceeded.
5. Confirm the final balance after the adjustment status becomes POSTED.

The quick-check list below is intentionally formatted as ordinary text, not a data table:

- • Verify the selected warehouse.
- • Confirm the SKU matches the physical label.
- • Check that the adjustment reason is accurate.

4. API response semantics

Response	Meaning	Client behavior
INV-200	Adjustment posted	Refresh stock balance and show adjustment ID.
INV-202	Approval required	Show pending approver and prevent duplicate submission.
INV-409	Stock version changed	Reload latest balance and ask operator to review.
INV-422	Validation failed	Highlight invalid fields; keep entered values.

5. Configuration quick reference

Default approval threshold 25 units
Maximum bulk rows 250
Audit retention 365 days
Inventory timezone UTC

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7. Error handling

If the inventory service cannot be reached, show a retryable error and do not repeat a POST automatically unless an idempotency key was supplied. A response timeout does not prove that the adjustment failed; refresh the adjustment record before retrying.